

Inference $\rightarrow$ Prediction

Input $\xrightarrow{\text{(Time)}}$ Output

{ Model Inference }

Build Your LLM

* 1) Pre Training :- Scratch / Zero Training

* 2) Fine Tune :- Pre Trained / Base / Foundational
Domain specific Task

+

Domain specific Data

$\downarrow$

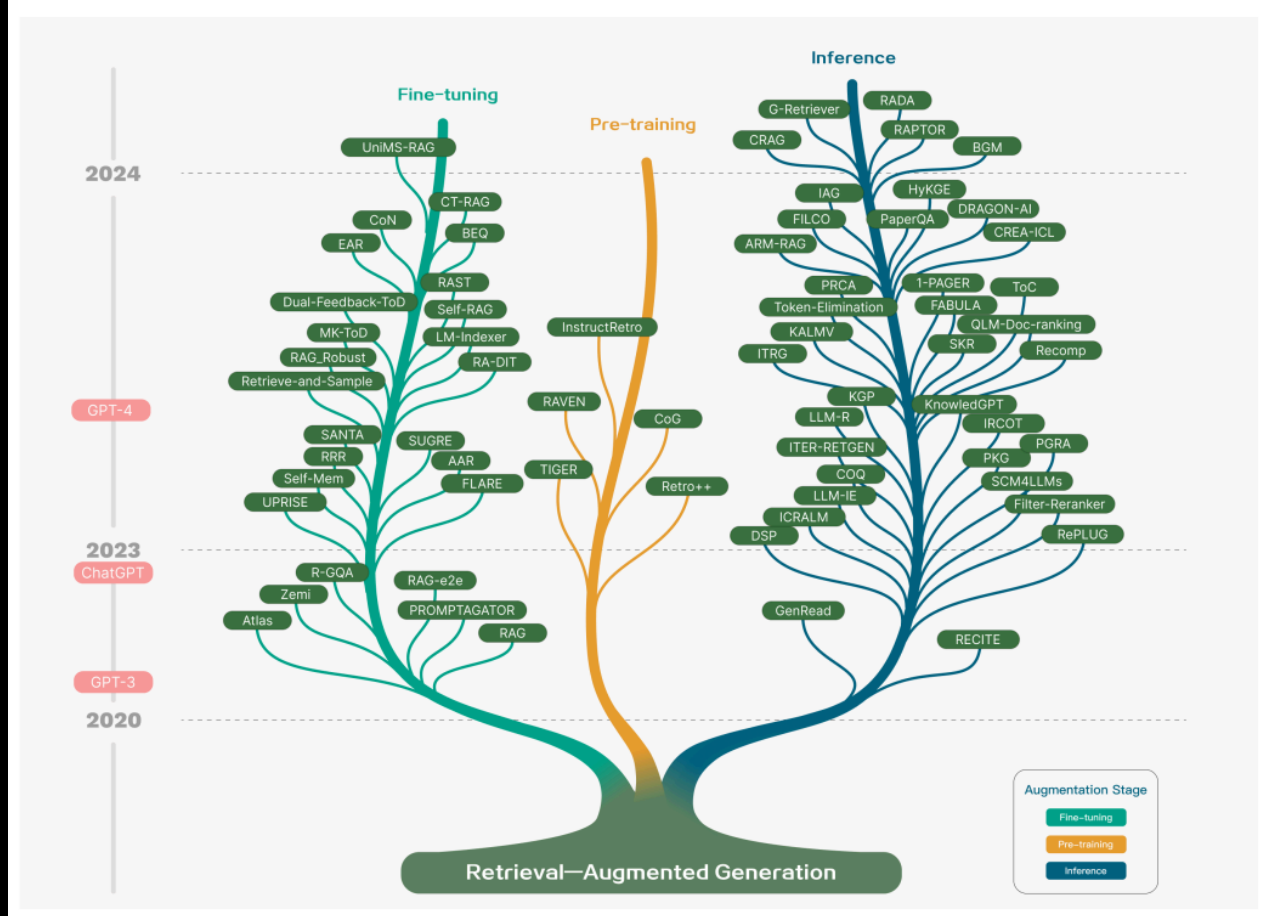
New Model

3) Inference :- Directly using a pretrained model
GPT, Claude, Cohere.

LLM :- Trained on some data Maths

Your data :- Not Relevant Bioinformatics

Fine Tune :-
* 10 million USD
R&D 1 million USD



Modes

PTM → CoG, TIGER (2025)

FTM → Self-RAG, FLARE (2025)

INF → CRAG, Iter-RETGEN

Papers

1) Pretraining → Transformers

1) REALM → 2 (LLM + Retriever)

2) RETRO → Retrieval based Transformer Enhanced

3) Atlas → Few Shot with Retrieval
Few Examples → LLM Prompting

2023
CoG → Copy Generation

2024
* Tique :- Most Advanced Pre-Trained
RAVEN :- Retrieval Augmented Encoder Decoder

Fine Tuning

1) Self - RAG

2) FLARE :- Forward looking Active Retrieval

3) UNiMS :-

4) Self-Mem :-

Query Optimization

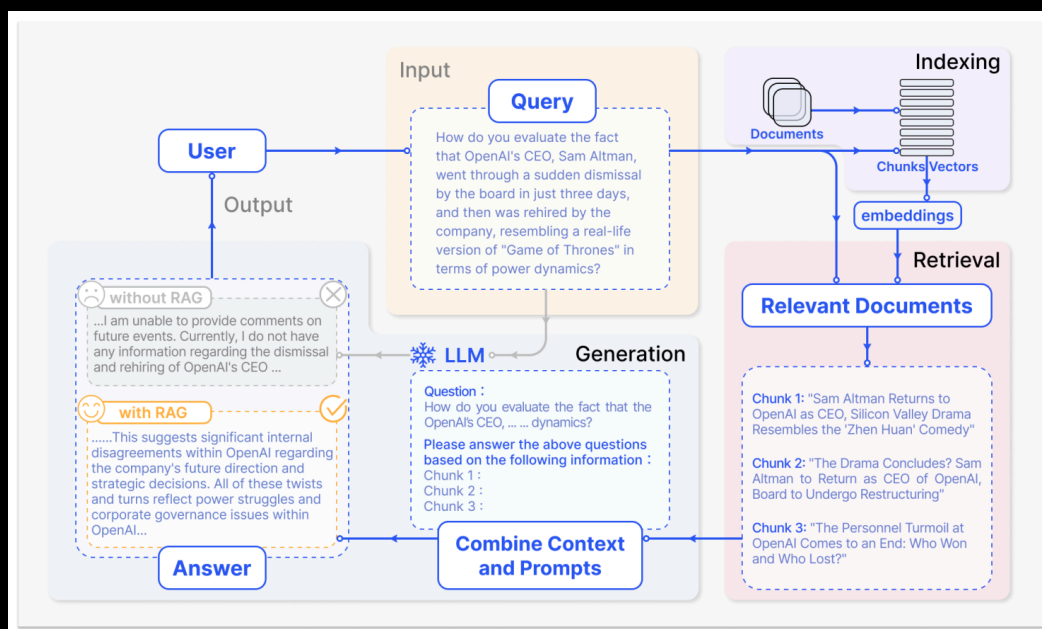
HYDE :- Hypothetical Embeddings

RAPTOR :- Tree based Retrieval

GRAPH-RAG

* CRAb :-

* Iter GEN :-



Phase:-

Data sources

↓
Chunks → vectors → Vec DB

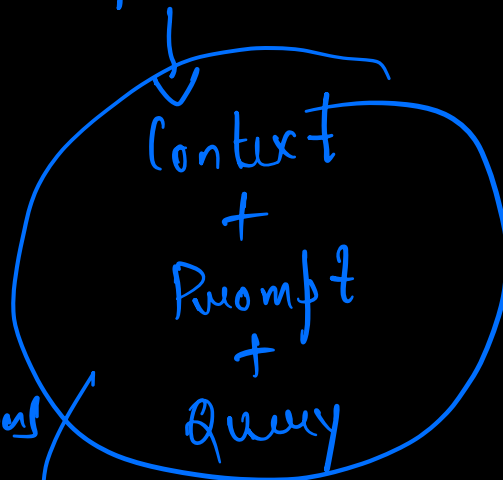
Query:- Question

Prompt:- Set of Instructions

Phase:-

User → Query → Embedding

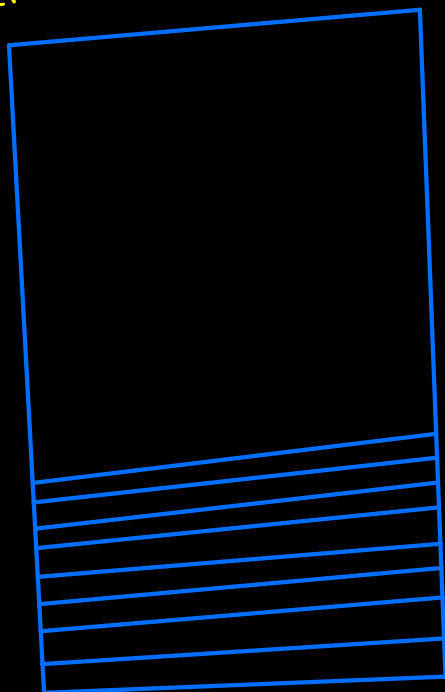
TopK chunks ← Similarity



LLM → generation

→ Response

Input



Output

Model :: 200 layers

Fine Tuning

150 layer - Frozen

(Weights + Biases)
Don't change

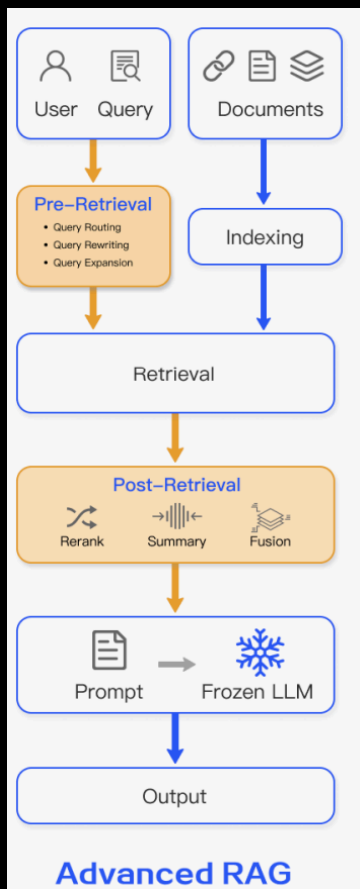
50 - Retraining
(w & B will change)

Specific Data

Naive RAG

- 1) Linear & Sequential
- 2) Retrieval Read Pattern
- 3)

Advanced RAG



i) Pre Retrieval

(i) Query Routing :- Directing to data source

(ii) Rewriting :- Reformulate / Rephrase

(iii) Expansion :- Adding related related terms.

(Unclear question)
(I would) the

Parametric (LLM) + Non Parametric (Retriever)

Rewriting : { Q1 | Q2 | Q3 }
 Response to be exactly correct

Post Retrieval Top (K) Chunks
 (operation) ^{5, 10, 15}

- (i) Rerank :- Reorder by Relevance
 * Reranking Models * Chunk 1
2
3
4
5
- (ii) Summary :- Compress retrieved content
Chunk 3
4
1
2
5
- ✓ Chunk 1 :- 128
 - ✓ Chunk 2 :- 128
 - ✓ Chunk 3 :- 32
- Summarized Retrieved Content
 (Info loss) →

* Concatenation *

(Maintaining shorter context window size)

LLM :- Context Size ↑ { Response Time Increase }

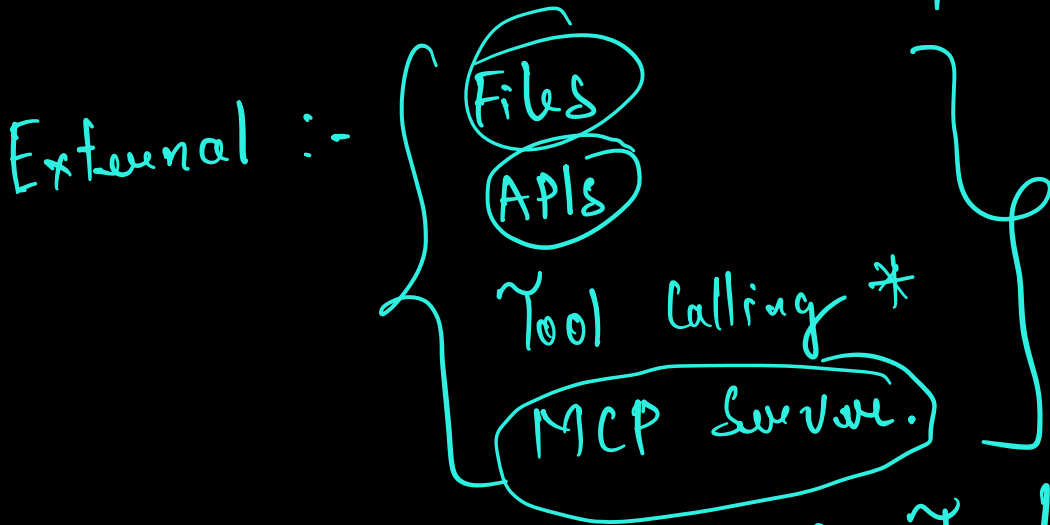
Token Consumption Increase



Price Increases

Fusion:-

Combine information sources multiple.



RAG + Google Search Tool

↓
Latest Event

Query Decomposition

(Recursive)

1) **Actual**
How does the GDP of countries that hosted Olympics after 2000 compare?

- D1 Q1) Countries hosted Olympics after 2000
Spain, Greece, Australia.
- D2 Q2) GDP of Spain
- D3 Q3) GDP of Greece
- D4 Q4) GDP of Australia
- D5 Q5) Compare all of them.

Adaptive

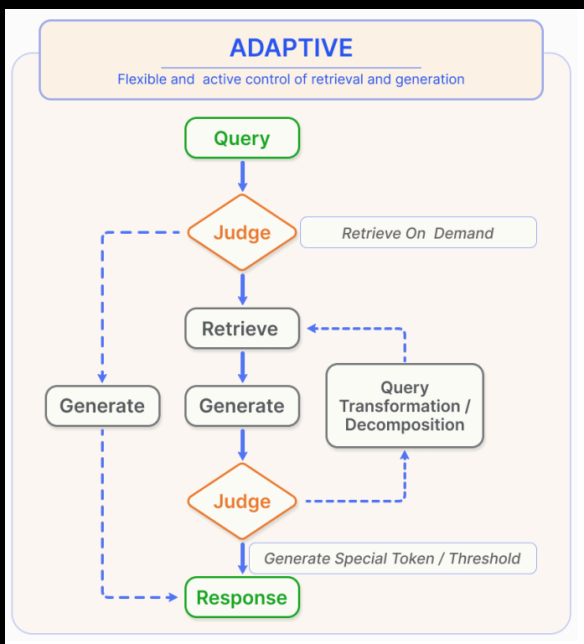
Judge output :- threshold confidence

if confidence (generated) < threshold
trigger retrieval

elif contain [TOKENS]
retrieval & continue

else generation

[Active] → Reflection Tokens



Self-Reflect

[RETRIEVE], [ISUSE], [CONTINUE]
Reflexion Tokens.